

Context Inheritance: Capability-Dependent Benefits of Repository Context for AI Coding Agents

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ABSTRACT

Giving an AI coding agent a repository’s operating context—its decisions, constraints, and conventions—sometimes helps a great deal and sometimes not at all. We show the difference is governed by a single variable: the capability of whoever *wrote* the context relative to the agent that *reads* it. Across pre-registered experiments on SWE-bench Verified spanning a small local executor (Gemma 3, 8B) and a frontier executor (Claude Opus 4.8), a repository context layer raises a weak reader’s gold-file localization by 26 percentage points ($p < 0.0001$) when the context was authored by a more capable model, and by exactly nothing (+0.0 pp, $p = 1.0$) when the reader authored it itself. The benefit is a capability *threshold*, not a dose curve: any author above the reader roughly doubles the reader’s localization; a self-taught weak model equals no context. At the top of the range the picture inverts along a second axis—*what* transfers. A frontier agent gains from its own failure-derived lessons (+14.6 pp resolve on constraint-sharing tasks, $p = 0.041$; −32% tokens per resolved task) but nothing from answer-derived summaries, while a weak model benefits most from exactly those summaries. On a broad random sample the frontier agent sits at a resolve ceiling (91% with no context) that no delivery mechanism improves, and where reading a context store only adds token cost. We call the organizing principle *context inheritance*: repository context is senior knowledge that propagates *down* a capability gradient. The practical rule follows directly—seed and maintain context layers with the most capable available author, and expect the benefit to accrue to the cheaper agents that read them.

KEYWORDS

AI coding agents, repository context, capability gradient, knowledge transfer, SWE-bench, pre-registered study

1 INTRODUCTION

A modern coding agent, handed a bug, can plan, edit, run tests, and iterate for hours. What it cannot do is know the things a repository never wrote down: which library the team removed and why, which layering rule is unwritten, which approach looks reasonable but was tried and rejected. A *repository context layer* [1, 2]—a small, version-controlled, human-readable store of a project’s operating context, consulted by the agent before it acts—is one answer to this gap. But a practitioner who adopts one quickly meets a confusing

fact: sometimes it transforms the agent’s behavior, and sometimes it does nothing at all.

This paper explains the variance. We find that the benefit an agent draws from repository context is *capability-dependent*: it is a function of how capable the context’s author was relative to the agent reading it. When a stronger author writes for a weaker reader, the reader inherits a large, statistically decisive navigational benefit. When author and reader are the same weak model, the benefit is exactly zero. And when the reader is already a frontier model operating near its ceiling, generic context adds no resolution and only spends tokens.

We establish this across a family of pre-registered experiments on SWE-bench Verified [3], varying three factors independently: the *executor* (a small local model, Gemma 3 8B, and a frontier model, Claude Opus 4.8); the *author* of the context (none; the executor itself; a more capable model writing blind; and a labeled oracle bound); and the *delivery* mechanism (injected into the prompt, a monolithic file, or a sharded store the agent reads selectively). Our contributions are:

- (1) **A capability gradient in context benefit.** With the executor held fixed at 8B, a blind frontier-authored seed raises gold-file localization from 21.1% to 47.2% (+26.1 pp, $p < 0.0001$), while the executor’s own promoted lessons move it not at all (+0.0 pp, $p = 1.0$). The effect is a threshold: context helps iff its author outranks its reader (Section 4).
- (2) **An inversion in what transfers, by reader capability.** A weak reader benefits most from answer-derived summaries (where things are); a frontier reader benefits only from its own failure-derived lessons (where it will go wrong), gaining +14.6 pp resolve on constraint-sharing tasks ($p = 0.041$) (Section 5).
- (3) **A frontier ceiling, and the cost of context there.** On a broad random sample the frontier executor resolves 91% of well-specified bugs with no context; no delivery mechanism improves this, and the selective store costs more tokens, not fewer, at high capability (Section 6).
- (4) **Token economics of inheritance.** Where context does help a capable reader, it pays for itself: −32% tokens per resolved task (Section 7).

The synthesis is a design principle we call *context inheritance*: repository context behaves like senior engineering knowledge, and its value propagates *down* a capability gradient—from a strong author, once, to every cheaper agent that later reads the repository.

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2 BACKGROUND

Teams already convey project knowledge to agents through instruction files (CLAUDE.md, AGENTS.md), IDE memories, and retrieval-augmented generation. A repository context layer differs in being co-versioned with the code, human-reviewed, and consulted by contract rather than by retrieval similarity; the architecture and its properties are specified elsewhere [1, 2] and are not the subject of this paper. Our question is narrower and empirical: *given* such a layer, what determines whether an agent benefits from it? Prior agent-memory work (Reflexion, Voyager, MemGPT, Generative Agents [4–7]) studies a single agent accumulating its own experience. The phenomenon we isolate is cross-capability: knowledge written by one model and read by another, of different strength.

3 METHOD

The protocol (hypotheses, conditions, metrics, analysis plan, power simulation) was pre-registered and externally timestamped at a git tag before any confirmatory run; pilot runs are excluded from confirmatory analysis. All materials—the frozen protocol, the blind-authoring audit, episode transcripts, promotion logs, and containerized evaluation verdicts—are released for replication [8].

Executors. A small local model (Gemma 3, 8B, quantized) and a frontier model (Claude Opus 4.8), each run in an identical minimal agent scaffold: read files, edit code, run individual tests, up to a fixed turn budget, leaving an uncommitted working-tree patch. Every episode is a fresh model session in a freshly cloned checkout; no state carries between episodes.

Context authors. Context is *none*; authored by the executor itself through a consult–execute–learn–promote lifecycle (self-authored); authored by a more capable model given only a years-old, history-stripped checkout and constraint-group names, never a problem statement, patch, or test (a blind seed); or authored with sight of the maintainers’ gold patch (a labeled oracle bound, never pooled with confirmatory arms). Crucially, **no context was produced by an automatic extraction pipeline**: all authored context is blind or lifecycle-derived by design, so that the study isolates the capability-of-author effect and cannot leak held-out solutions. Automatic extraction is deferred to future work (Section 9).

Delivery. Where context is present it is delivered as an injected prompt block, a single monolithic context.md the agent is instructed to read, or a *sharded* store (an index plus one file per decision) the agent reads selectively under a consultation contract.

Instances and metrics. Two instance regimes are used and never conflated: ordered *transfer pairs* (A, B) within a repository whose fixes depend on the same constraint (the regime the layer is designed for, about 8% of Verified), and a broad stratified random sample. Metrics: *consultation* (did the agent read the context, from transcripts), *gold-file localization* (did it edit a file the maintainers’ fix touched), *resolve* (did its patch pass the hidden test suite, via the official SWE-bench containers), and *tokens* per episode and per resolved task. Paired comparisons use sign-flip permutation tests (10k draws, fixed seed); families are Holm-corrected.

Table 1: Weak executor (8B), held-out localization by context author. 180 episodes/arm.

context author	non-empty	gold-hit	resolve
none	68.3%	21.1%	5.6%
self-authored (8B)	—	21.1%	—
blind frontier seed	81.1%	47.2%	6.1%
gold-distilled (oracle bound)	79.4%	42.8%	7.8%

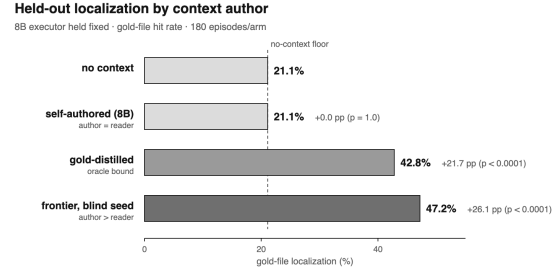


Figure 1: Held-out gold-file localization of the 8B executor by context author (36 pairs $\times$ 5 repetitions). The dashed line is the no-context floor; paired deltas versus that floor are annotated. Self-authored context sits exactly on the floor; any author more capable than the reader roughly doubles localization. Resolve-rate deltas are not significant at this tier.

4 THE CAPABILITY GRADIENT

Holding the executor fixed at 8B (36 transfer pairs $\times$ 5 repetitions = 180 held-out episodes per arm), the held-out outcome varies only with who authored the context (Figure 1).

A blind frontier-authored seed raises gold-file localization from 21.1% to 47.2%—+26.1 pp, paired $p < 0.0001$, far beyond the pre-registered minimum detectable effect. Gold-distilled context, an upper bound, raises it to 42.8% (+21.7 pp, $p < 0.0001$); the two capable-author tiers are statistically indistinguishable. The executor’s own promoted lessons—162 of them across 60 pair-repetitions—move localization by +0.0 pp ($p = 1.0$): exactly the no-context control.

Two readings are ruled out. This is not a dose curve in which more capable authorship yields proportionally more benefit; blind-seed and oracle are a tie, so the shape is a *threshold*—context helps if and only if its author outranks its reader. And it is not that the 8B model cannot use context: it uses frontier-written context decisively; it simply cannot *write* context worth reading. Inspection of the self-authored store shows why: the weak model records tooling tactics (“run pytest with -x”), not repository constraints, whereas the frontier author, under the identical learning prompt, records the constraint-and-localization facts the transfer clusters center on.

At this tier the movement is in localization and episode completion, not resolve: all arms sit at 5–8% resolve, and no resolve claim

Table 2: Frontier executor (Opus 4.8), broad 88-instance sample. Consultation is the fraction of episodes that read the context; cost is notional, at list prices, excluding prompt caching.

delivery	resolve	consult	prompt-tok	\$/resolved
none (B)	90.9%	0%	25.8K	0.153
injected (E)	81.8%	0%	23.5K	0.157
file (FILE)	85.2%	100%	24.0K	0.154
sharded (SHARD)	88.6%	100%	38.4K	0.230

is supported (the pooled oracle delta is +2.2 pp, $p = 0.22$). Localization is a proxy for working in the right place, not for correctness; we report it as such.

5 WHAT TRANSFERS DEPENDS ON THE READER

The gradient says a capable author helps a weak reader. At the top of the range, a second axis appears: *what kind* of knowledge transfers depends on the reader’s capability.

Running the full consult–execute–learn–promote lifecycle with the frontier executor on 16 constraint-sharing pairs ($\times 3$ repetitions, 48 held-out episodes/arm), the agent’s own failure-derived lessons—extracted from working task *A*, never seeing a gold patch or hidden test—raise held-out resolve on task *B* from 58.3% to 72.9%: +14.6 pp, paired $p = 0.041$, the frontier-tier pre-registered primary. Yet lessons distilled from the *gold patch* of task *A* transfer nothing at this tier (60.4% resolve, indistinguishable from no context).

This is the exact inverse of the weak tier, where gold-distilled context was the *strongest* author. The interpretation is sharp: answer-derived lessons encode solutions; failure-derived lessons encode traps and process. A weak model needs to be told where things are; a strong model already localizes and instead needs to be told where it will go wrong. For context-layer design this means the transferable payload changes with the reader—solution summaries for weak readers, process and failure knowledge for strong ones.

6 THE FRONTIER CEILING

Both results above use constraint-sharing pairs, the regime a context layer is built for. What happens when a frontier agent meets a *broad* random sample with generic repository context? We ran all four delivery conditions—no context (B), injected (E), monolithic file (FILE), and sharded store (SHARD)—with the frontier executor on an 88-instance stratified subset of Verified, one repetition (Table 2).

Two facts stand out (Figure 2). First, the no-context baseline already resolves 90.9% of these well-specified, isolated bugs: the frontier model does not need repository context to fix bugs it can already localize, and there is no headroom for context to add resolution. No delivery mechanism beats the baseline; the small differences among B/E/FILE ($\pm$ a few instances on $n = 88$) are within noise. Second, the consultation contract binds perfectly—FILE and SHARD read the context in 100% of episodes—so the null is not a failure to read; it is that reading generic context on a random bug has nothing to contribute.

The frontier ceiling: broad-sample delivery matrix

Opus 4.8 · 88 random instances · resolve rate; no delivery beats no-context

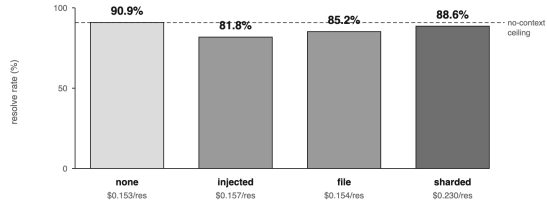


Figure 2: Frontier executor on a broad 88-instance random sample: resolve rate by delivery mechanism. No context arm exceeds the no-context ceiling (dashed); the selective store (sharded) recovers most, but at the highest per-resolved cost. Costs are notional, at list prices, excluding prompt caching.

The ordering among the context arms is nonetheless consistent with the weak-tier finding that *selection beats volume*: injecting the whole payload (E, 81.8%) hurts most, selective reading (SHARD, 88.6%) hurts least, monolithic reading (FILE, 85.2%) sits between. But at frontier capability the selective store’s cost *inverts*: a capable model turns “read only what you need” into thorough exploration, so SHARD spends 38.4K prompt tokens against FILE’s 24.0K and costs ~50% more per resolved task. At the weak tier the same sharded store was the *cheapest* arm, paying roughly three times fewer context tokens than the monolith. Selection is economical for readers that read sparingly; a strong reader reads thoroughly.

Together with the gradient, this completes the capability story: context benefit requires an author above the reader *and* a reader below its ceiling. A frontier model on a broad, well-specified bug set violates the second condition, and generic context becomes a cost rather than a benefit.

7 TOKEN ECONOMICS

Where context does help a capable reader—the constraint-sharing regime of Section 5—it is self-financing (Figure 3). The self-taught frontier lifecycle spends 55.7K tokens per resolved task against 81.8K with no context: –32%. The decomposition is instructive: episodes that end in a fix cost about the same in both arms (~33.6K); the savings come entirely from fewer doomed explorations. Context of any kind shortens wandering (the oracle arm saves tokens identically), but only failure-derived context also converts the saved effort into additional resolves. For a strong reader in the right regime, context is paid back in tokens; for a weak reader it is a paid navigational upgrade (the injected block costs +20% tokens per episode even as it shortens episodes).

8 DISCUSSION: CONTEXT INHERITANCE

The four results resolve into one principle. Repository context is not a uniform good that any agent draws on equally; it is *senior knowledge*, and its value flows down a capability gradient. A reader inherits benefit in proportion to how far its context’s author outranks it, and only while the reader has headroom to improve. This reframes several practical questions:

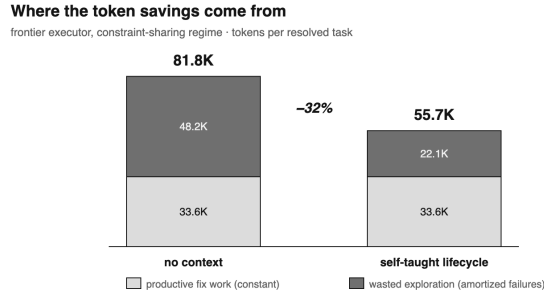


Figure 3: Frontier executor, constraint-sharing regime: tokens per resolved task, decomposed into productive fix work (constant, ~33.6K) and wasted exploration amortized from failed episodes. The ~32% saving comes entirely from the shrinking exploration term, not from cheaper fixes.

Who should write the context? The most capable author available—a human expert, a frontier model, or distillation from authoritative fixes. A project should not rely on its cheap working agents to author their own context; a weak self-taught loop equals no context exactly.

Who benefits? The cheaper agents. One frontier authoring pass, paid once, propagates senior-level navigational knowledge to every local agent that later reads the repository at zero marginal inference cost. This is the economic case for the pattern: expensive knowledge, written once, read cheaply forever.

When is it worth it? When the reader is below its ceiling on the task distribution—weaker models broadly, and strong models on tasks that are ambiguous, convention-laden, or re-hit a known constraint. On well-specified bugs a frontier model already resolves, generic context is not worth its token cost.

9 THREATS TO VALIDITY

Benchmark scope. SWE-bench Verified supplies well-specified, isolated bug fixes graded by hidden tests. It cannot measure a context layer’s central real-world value—adherence to conventions and avoidance of rejected approaches, where “correct” is not “tests pass.” The frontier ceiling of Section 6 is therefore a statement about this task type, not about repository context in general; a benchmark that scores convention-adherence is needed to probe the value the frontier ceiling hides, and is the natural next instrument.

Extraction untested. All authored context here is blind- or lifecycle-derived, deliberately, to isolate the author-capability effect and guarantee no leakage of held-out solutions. Whether an automatic extraction pipeline can *produce* context of comparable value is a separate question this design brackets; it is the clearest follow-up—vary the context’s *author* (blind-frontier vs. machine-extracted vs. human-curated) with delivery and executor fixed.

Statistical scope. The frontier resolve effect ($n = 16 \text{ pairs} \times 3$) and the broad-sample matrix ($n = 88$, one repetition) have wide intervals; the gradient and its threshold shape, pooled to 180 episodes/arm, are the robust core. Contamination affects both arms of every within-model comparison symmetrically, compressing deltas toward zero.

Transfer pairs target constraint-shaped bugs by construction—construct validity, since recurring constraints are the only setting in which shared context can pay off.

10 CONCLUSION

Whether a repository context layer helps an AI coding agent is not a property of the layer alone; it is a relation between the capability of the context’s author and the capability of its reader. A stronger author lifts a weaker reader decisively—doubling its localization—while a self-taught weak model gains nothing and a ceilinged frontier model gains nothing generic delivery can add. Context flows down the capability gradient. The design consequence is concrete: author repository context with the strongest writer you have, and reap the benefit in every cheaper agent that reads it.

CONFLICTS OF INTEREST

All authors are affiliated with Metatron Research, whose open-source implementation of the repository-context pattern (Metatron) motivates this study. The experiment mitigates this competing interest by pre-registration before data collection, a scaffold-uniform minimal harness rather than the product itself, context that is blind- or lifecycle-authored rather than produced by that implementation, and fully public data and analysis code.

AI-ASSISTED PREPARATION STATEMENT

Consistent with COPE guidance, the authors disclose that large language models (Anthropic Claude) were used to assist with manuscript drafting, analysis scripting, and experiment orchestration, under continuous author direction and review. The authors take full responsibility for the final text and all reported results.

DATA AVAILABILITY

The frozen pre-registration, blind-authoring audit, episode transcripts, promotion logs, containerized evaluation verdicts, and analysis scripts are released for replication [8].

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